Venkata Saratchandra Patnaik Markonda Patnaikuni

+1 6023018122 | sarath92015@gmail.com| https://www.linkedin.com/in/saratchandra-92015/

EDUCATION

Master of Science in Computer Science (CGPA: 3.78/4) **Expected December 2025**
Arizona State University, Tempe, AZ.
Bachelor of Technology in Computer Science (CGPA: 9.11/10) **May 2022**
Christ University - Faculty of Engineering, India.

TECHNICAL SKILLS

Programming Languages: Python, Kotlin, Go, JavaScript, TypeScript, C, C#
Web & Frontend: React.js, Node.js, Express.js, HTML, CSS, Bootstrap, Android Studio
Databases: MySQL, PostgreSQL, Firebase, MariaDB/ MongoDB
DevOps & Cloud: AWS (S3, EKS, EC2, Lambda, Load Balancer), Docker, Kubernetes, Jenkins, REST APIs
Development Practices & Version Control tools: Extreme Programming (XP), Agile, Pair Programming, GitHub

PROFESSIONAL EXPERIENCE

Implementation Engineer: Amagi Media Labs, Bengaluru, India **August 2022 – November 2023**

- **Solution Architecture & Automation:** Designed onboarding solutions for content creators, **optimized video ingestion and distribution**, and automated customer onboarding using **Terraform**, reducing manual setup time by **90%.**
- **Cloud & Deployment:** Managed **15+** containerized microservices on **AWS EKS with Kubernetes and Docker**, automating deployments with Linux scripting, and improving **weekly deployment frequency** by **30%**.
- **Scripting & Content Delivery:** Developed **Python** scripts for video preprocessing, **reducing processing time** by **37%**, automated **API**-based video access verification, and ensured global content delivery via CDNs, reducing failures by **95%**.

Web Development Intern: Blueplanet Solutions Inc, India **April 2021 - June 2021**

- Engineered a **MySQL database** with **stored procedures** for 'Campus Club,' accommodating **1,000+ student profiles** and enabling efficient data retrieval, reducing query response times by **60%,** and enhancing user experience.

PROJECTS

Underground Cable Fault Detection (IoT)

- Developed an **IoT-based** system using **Arduino** and **8051 microcontrollers** in **Proteus** to detect short-circuit and open-circuit faults in underground cables, utilizing a potential divider network to monitor voltage changes.
- Implemented real-time fault localization, displaying fault locations (e.g., 2 km, 4 km, 6 km) on an LCD, reducing manual inspection effort and cost by enabling precise fault detection over the internet.\
- Report ⟶ https://drive.google.com/file/d/14ajvIOz8b-suEc9X70d4L_8_En7kqm69/view?usp=drive_link
- Demo Video ⟶ https://drive.google.com/file/d/1ThyB2oyNVQNJAAw4KQEddeaqQbcosGI5/view?usp=drive_link

Handling Imbalanced Data on Bank Churn Prediction (Machine Learning & Data Analytics)

- Developed a machine learning model to predict customer churn using **Python**, addressing imbalanced datasets with **SMOTE** and **undersampling**, improving model accuracy by 15%.
- Performed data cleaning and visualization with **Pandas** and **Matplotlib**, extracting key features for classification that were relevant to multimodal data processing in vision research

Automated Online Attendance Bot (Automation Anywhere)

- Automated Webex **attendance tracking** using an **RPA bot** built with Automation Anywhere A2019, eliminating **95% of manual data entry** and reclaiming 10 hours per week for the team.
- Implemented a **50-minute threshold** for accuracy, reducing manual effort and improving efficiency.